→ Natural language Processing
→ Basics
→ for Generative AI.

What is NLP?
Ans: NLP is a field at the intersection of linguistics, Computer Science, and Artificial intelligence that focuses on enabling machines to understand, interpret, generate, and respond to human language.

Key concepts of NLP.

① Text Processing:
Cleaning and Converting raw text into a form machines can work with.

② Tokenisation:
Splitting texts into words, subwords, or sentences.

③ Stop word removal:
Removing common words like (is, the) that dont add meaning

④ Stemming and lemmatisation
Reducing words to their root form (eg running → run)

⑤ POS tagging:
Identifying grammatical parts of words (noun, verb etc)

⑥ Named entity recognition (NER)
Detecting names, place, dates, etc from text.

⑦ Dependency parsing
Finding how words related to each other in a sentence.

→ Why NLP is essential for Generative AI?.
Ans: Gen AI like (Chat GPT and Gemini etc) generates human like texts. for this, it needs a deep understanding of NLP tasks to.

→ Understanding Context
→ Maintain grammer and flow
→ Know intent and emotion

→ Performs text Generation, summarization, translation, question answering etc.

→ Core NLP techniques and Concepts to learn.

① Bag of words (BOW)

Ans: Represents texts as a set of word count
eg.
"I LOVE NLP and NLP is love"
→ {I : 1, LOVE : 2, NLP : 2, and : 1, is : 1}

② TF-IDF :
Highlights important words by reducing weights of common ones.
eg. word "the" gets low score, "NLP" gets high.

③ Word Embedding :
Words converted to dense vectors that capture meaning.
eg.
Similar words → closed vectors eg. word2vec, Glove.

④ Transformer :
Deep learning model uses attention mechanisms to understand sequence of words.

eg. Foundation of models like BERT, GPT

⑤ Attention mechanism :
Tells the model which words in a sentence to "focus" on.
eg.
In translation, focus on Subject while generating verb.

⑥ Syntax :
The structure and grammatical rules of language (eg. words in sentence).

⑦ Semantics:
The meaning of words and sentences, and how many changes with context.

⑧ Pragmatics:
Understanding language in context, including speaker intent and situational meaning.

⑨ Discourse
Analysis of language beyond single sentences, focusing on coherence in conversational or texts.

→ Fundamentals of NLP techniques.

Tokenisation → Stemming and lemmatization → POS tagging → Named entity recognition
↓
Sentiment analysis → Parsing → Stop word removal.

→ NLP workflow:

① Data preprocessing:
Clean and prepare texts using tokenisation, stemming, lemmatization, and stop word removal.

② Feature extraction:
Converts text into numerical representations (eg word embedding like word2vec

③ Modeling:
Use algorithm (logistic regression, Naive Bayes, neural networks, transformers) to analyze or generate language.

④ Evaluation:
Assess model performance with metrics like Accuracy, precision, recall, and f1 score.

Generative AI:

① LLM: large language models.
LLM's are the specific type of Neural networks that focus on understanding Natural language.

→ LLM's learns by reading books, texts (tons of data)

* How LLM are different

traditional programming

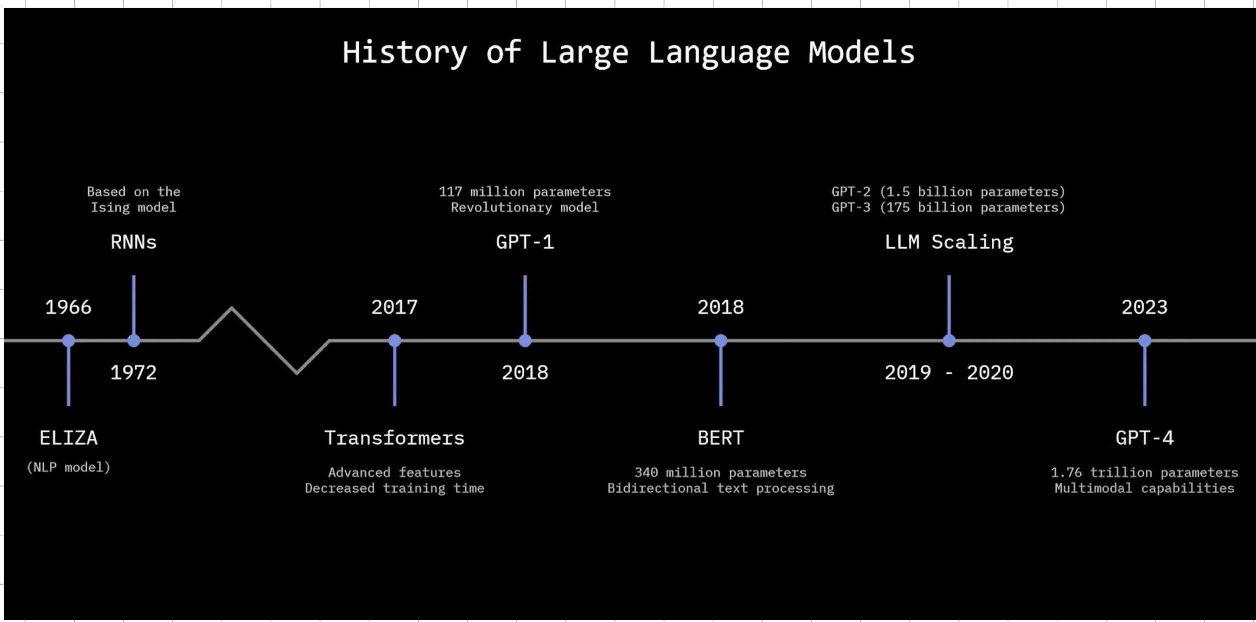
Its instruction based.
if x then y
if this then that

LLM's

It's teaching how to learn how to do things.

→ LLM's are much more flexible, adaptible, accurate
→ Text Summarisation
Text Generation
Creative writing.
Question and Answer.
Programming.

→ History of LLM.



* How LLM's works.

There are 3 steps.

① Tokenisation: It Splits long texts into individual tokens.
eg.
what is the tallest Building ?

Tokenisation → what is the tall est Building

Token 1 Token 2 token 3 token 4 token 5 token 6

This step is done so that model learns each step individually. and as grouping of words.

② Embeddings :

→ Embedding are the numerical representation of words, phrases, or even entire texts that capture their semantic meaning.

→ These embeddings allows LLM's to understand the relationship b/w different pieces of texts, making them more effective at tasks like translation, question answer, and content generations.

③: Transformers:

Transformers refers to Deep learning architecture that serves as the foundation for how these models process and generate human like texts.

→ A Transformer

- Understands the Context of a words more effectively via attention mechanism.
- Processes sequences in parallel. (Unlike older models like RNN's / LSTM that worked sequentially).
- Scale efficiently for training massive dataset.

Key Components of Transformer.

ⓐ Embedding layers:
Converts input tokens (words or sub words) into higher dimensional vectors.

ⓑ Positional encoding:
Adds information about the position of each words in a sentence, since transformers process input in parallel and not sequentially.

ⓒ Multi head self attention:
- Allows the model to weigh the importance of each word in relation to every other word in the sequence.
- Example: In the sentence "The Animal didn't cross the road because it was too tired",
  attention helps the model to understand that "it" refers to "the animal".

ⓓ Feed forward layers:
fully connected layers applied after the attention layers to transforms the representations.

ⓔ Layer Normalisation and Residual Correction:
Helps stabilise and speed up the training.

———x ———x ———x ———x ———

Summary of 1st unit:

Concept	Key points
NLP	→ Enables computers to understand language
N-gram model	→ Fixed content, limited understanding
RNN/ LSTM	→ Sequential, better context, but slow
Transformer	→ uses attention, parallel, handles long-range dependencies
Self attention	→ learns importance of each word in context of others.

In order to understand Gen AI, you need to understand

Word embedding:

→ representing each words in a form of vector.

	battle	horse	king	man	queen	..	woman
authority	0	0.01	1	0.2	1		0.2
event	1	0	0	0	0		0
has tail?	0	1	0	0	0		0
rich	0	0.1	1	0.3	1		0.2
gender	0	1	-1	-1	1		1

dimensions.

$$\underset{\text{King}}{\begin{bmatrix}1\\0\\0\\1\\-1\end{bmatrix}} - \underset{\text{man}}{\begin{bmatrix}0.2\\0\\0\\0.3\\-1\end{bmatrix}} + \underset{\text{woman}}{\begin{bmatrix}0.2\\0\\0\\0.2\\1\end{bmatrix}} = \begin{bmatrix}1\\0\\0\\0.9\\1\end{bmatrix} \sim \underset{\text{Queen}}{\begin{bmatrix}1\\0\\0\\1\\1\end{bmatrix}}$$

→ Google word2vec Model has 300 Dimension.

→ No one knows what are the dimension

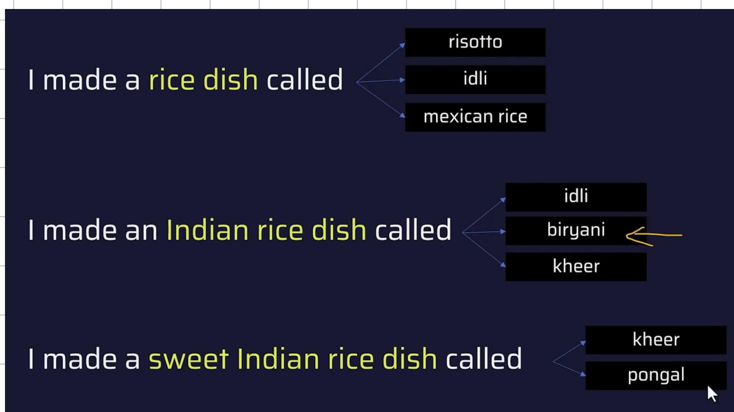
→ GPT has more than 12000 dimension. This is how the chatGPT is able to understand.

→ Word2Vec has limitation as the same word can mean different meaning with 2 sentences.

eg. He swung the bat and hit it for six
A bat flew from the Sky/Cave

Here the bat is refering to a different meaning.

Hence to overcome this problem, Contextual embedding came into picture.



I made a sweet Indian rice dish called kheer

$$\underset{\text{dish}}{\begin{bmatrix}1.09\\-0.72\\0.66\\\dots\\-0.8\end{bmatrix}} + \underset{\text{rice ness}}{\begin{bmatrix}-0.43\\0.91\\0.18\\\dots\\-0.75\end{bmatrix}} + \underset{\text{indian ness}}{\begin{bmatrix}0.29\\0.67\\0.52\\\dots\\0.85\end{bmatrix}} + \underset{\text{sweet ness}}{\begin{bmatrix}0.39\\-0.11\\-0.98\\\dots\\-0.46\end{bmatrix}} = \underset{\text{dish}}{\begin{bmatrix}1.34\\0.75\\0.38\\\dots\\-1.16\end{bmatrix}}$$

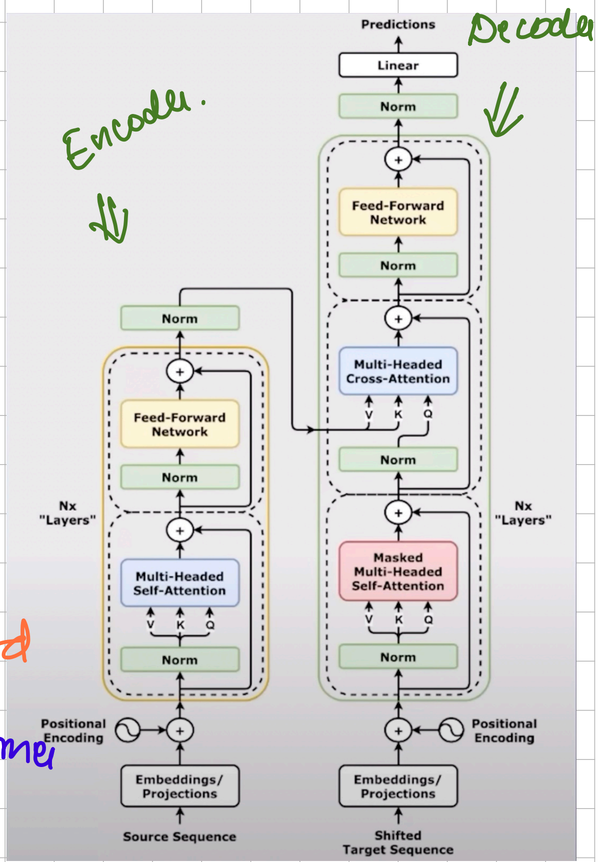
Static Embedding — Contextual Embedding

Static embedding: It assigns a single, fixed vector to each in the vocabulary, regardless of the context in which word appears.

Contextual embedding: Contextual embedding assigns different vector to a word depending on the surrounding words i.e the context.

Transformer Architecture

① The Architecture has 2 components encoder and decoder.



encoder: The purpose is to take the input sentence and generate the contextual embedding for each of the words or each tokens in that sentence.

Decoder: Once the contextual embedding is generated we feed that to a decoder and we try to predict the next word

when we talk about the transformer there are multiple task to be executed

ⓐ Next word prediction
ⓑ Translation

In terms of Translation, the decoder still produce Contextual embedding from encoder and you feed that to the decoder and decoder will start predicting the next word.
The encoder will produce the probability of the next word or the translation.

In case of next word prediction, the sentence is first passed through contextual embedding and in the decoder phase, the next word is predicted based on the probability.

The previous word is then again fed into the decoder phase to predict the next word.

The process is continued till the point the sentence is not completed.

→ The transformer model architecture is a generic architecture BERT and GPT are specific implementation based on transformer architecture.

→ The BERT Architecture only contains encoder part.
The BERT will only produce contextual embedding

→ The GPT has only decoder part, it takes the input and produce the output.

* A transformer model is a sequence to sequence model used in NLP tasks like translation, summarization, and more.

A model that translates or generates text by understanding the whole sentence at once - not word by words like RNN's.

① Input embedding:

→ Converts each tokens/words in the input sequence into a dence vector.

→ Input texts like raw texts like "I love NLP" is tokenised.

→ Each tokens are turned into a vector using learned embedding.

→ These are static sized numerical vectors.

② Positional Encoding:

→ It is needed because transformers dont understand order (position of words) by default.

→ Adds position information like 1st word, 2nd word ... to the embedding.

→ Uses sine and Cosine functions for each position.

"I love NLP" ≠ "NLP love I"

③ Encoder layer

Each encoder layer contains.

ⓐ Multi head self attention:

→ It Allows the model to focus on different parts of the sentence at the same time.

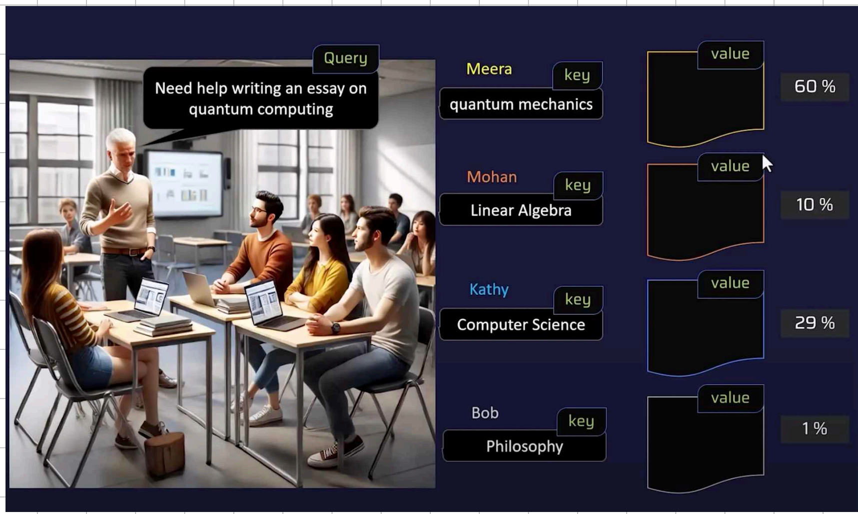
→ Attention helps the word "NLP" look at "I" and "love" to understand meaning.

→ Multi head means it learns from multiple perspective.

Inputs Q : Query
K : key
V : Value.

Example:
A professor wants to write a paper on Quantum Computing.

He then asked the student about their domain



Meera said : Quantum mechanics.

Mohan said : Linear Algebra

Kathy said : Computer Science

Bob said : Ph' ophy.

Here, query is professor want help in writing paper

Keys are : quantum mechanics
Linear algebra
computer Science
Philosophy.

Values are : The paper they write.

→ we can think of it as dot product. The dot product of Query and key is very good in case of Meera and Kathy.

ⓑ Add and Norm (Layer Normalisation + Residual)

→ Adds original input (skips connections) + normalise input
→ Help with faster training and better performance.

ⓒ Feed forward Neural network :
→ Applies 2 linear transformation with RELU activation in between.
→ Same FFNN applied to each position seperately.

→ This whole encoder block is repeated Nx times. 6 times in the original transformer.

④ Decoder layer :

Each layer has 3 sub layers:

ⓐ Masked multi head self attention

→ Prevents cheating - each word can only attend to earlier words, no future ones.

→ Example :
when generating "NLP", it can only see "I love", not what comes next.

ⓑ Multi head cross attention

→ Takes queries from the decoder, and keys and values from encoder output.

→ Connects source (input) to target (output).

ⓒ Feed forward Neural network.

Same as the encoder.

⑤ Final layer + softmax layer.

→ Converts decoder output into probabilities for each word in the vocabulary.

→ Example :
Outputs "translation" or "summary" word-by-word.

Summary :

① Embedding	Converts words → vector
② Positional encoding	Adds word order info
③ Self attention	Focus on important words
④ Multi head attention	look for multiple perspective
⑤ Masked attention	prevents looking ahead
⑥ Feed forward	Dense layer per word
⑦ Add & Norm	Normalise and stabilize
⑧ Cross attention	Connect encoder to decoder
⑨ Final Linear + Softmax	Output actual word.